

Performance Comparison: GPT-4.1 vs Claude

Head-to-head on Arbitration Intake:

- **GPT-4.1:** 3.08s, 3,370 tokens, 3 tool calls, **2,304 cached tokens**
- **Claude:** 9.96s, 5,695 tokens, 2 tool calls, no caching

GPT-4.1 is 3.2x faster and uses 40% fewer tokens — a dramatic performance advantage driven by prompt caching and more efficient token usage.

Two Workflow Patterns

1. Early Termination (Out of Scope, Vague Intake)

- Single iteration, no tools
- 3-7s execution
- ~1,500 tokens
- Agent quickly identifies invalid requests

2. Full Workflow (Arbitration, Employment, Conflict)

- 2-3 tool calls: `semantic_retrieval` → `conflict_check` (optional) → `case_routing`
- 10-13s execution
- ~5,300-6,400 tokens
- Complete intake processing

Claude Performance Across Scenarios

Scenario	Duration	Tokens	Tools
Arbitration	9.96s	5,695	2
Employment	12.72s	5,898	2
Conflict Flag	11.54s	6,366	3
Out of Scope	3.08s	1,486	0
Vague	6.63s	1,610	0

Key Findings

- ✓ Early termination pattern works well — invalid cases exit quickly
- ✓ GPT-4.1's prompt caching delivers massive speedup
- ✓ Claude's Arbitration trace **skipped conflict_check** (GPT ran it)
- ✗ No quality metrics/assessments attached to these traces

Recommendations

1. **Test GPT-4.1 across all scenarios** — current results show compelling advantages
2. **Investigate the missing conflict_check** in Claude's Arbitration trace
3. **Add quality scorers** to validate outcomes, not just performance
4. **Enable prompt caching for Claude** if you continue using it
5. **Profile tool latencies** to identify bottlenecks in the 10-13s full workflow cases

Explore GPT-4.1 trace Explore Claude Arbitration